

Required Assignment 2.2: AI and ML Use Cases — Techniques, Features, and Impact

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Use Case 1: Credit Scoring

Domain: Finance

AI/ML Applied Technique: Supervised Learning (classification / regression)

Data: Credit history, income - revenues, debt-to-income ratio, repayment and default history, employment history, existing obligations, transaction patterns based on historical evidence

Business Impact:

- Faster, more consistent lending decisions
- Lower default losses through better risk assessment
- Expanded access for thin-file customers using responsible alternative data

Reasoning and Explanation:

- Credit scoring is a supervised-learning problem because the model learns from historical applicant data where repayment or default outcomes are already known.
- Classification can predict whether a customer is likely to default or not.
- Regression can predict a continuous risk score or expected loss.
- The most relevant features are those that indicate repayment capacity and repayment intent.
- Key signals include credit history, income, debt-to-income ratio, repayment behavior, employment stability, existing obligations, and transaction patterns.
- The business value comes from faster decisions, more consistent underwriting, better risk pricing, and reduced default losses.
- Better scoring helps reduce bad debt while approving more good customers at the point of sale.
- **In AT&T's context, this applies to credit assessments for postpaid plans and device-installment financing.**

Use Case 2: Predicting Employee Attrition

Domain: HRM

AI/ML Applied Technique: Supervised Learning, binary classification

Data: Tenure, compensation details, market standards for compensation, performance ratings, promotion history, engagement survey scores, manager feedback

Business Impact:

- Proactive retention of critical talent
- Lower replacement cost and onboarding cost
- Targeted, timely interventions

Reasoning and Explanation:

- Attrition prediction is a supervised binary-classification problem because the model learns from historical employee outcomes: stayed or left.
- The model estimates the probability of a current employee leaving within a defined time period.
- Key features include tenure, compensation, performance, promotion history, engagement scores, absenteeism, manager, role, and commute distance.
- The business value shifts from reactive hiring to proactive retention.
- HR and managers can act early for high-risk, high-value employees.
- **In AT&T's context, this is useful for retaining scarce network and software engineering talent in business-critical roles.**

Use Case 3: Personalized Product Recommendations

Domain: Retail

AI/ML Applied Technique: ML Unsupervised Learning, collaborative filtering, Deep Learning, Neural recommenders

Data: Purchase history, browsing and clickstream behavior, product attributes, ratings and reviews, demographics

Business Impact:

- Higher conversion (Quote to Cash)
- Higher average order value
- Deeper customer engagement and retention

Reasoning and Explanation:

- Recommendation systems use unsupervised learning to find patterns across users, products, and behaviors.
- Collaborative filtering identifies similar users or products.
- Association rules identify products often bought or viewed together.
- Deep-learning recommenders capture richer customer preferences using embeddings and behavioral patterns.
- Business value comes from showing customers more relevant products with less effort.
- **In AT&T's context, this can support recommendations for devices, rate plans, accessories, and add-ons across digital and retail channels.**

Use Case 4: Email Spam Detection

Domain: IT

AI/ML Applied Technique: Supervised Learning, Text classification, Deep Learning, NLP / transformers

Data: Email text and tokens, sender reputation, header metadata, embedded links and attachments, send frequency and volume

Business Impact:

- Stronger security and phishing defense
- Less wasted employee time
- Protected brand and customer trust

Reasoning and Explanation:

- Spam detection is a supervised text-classification problem because the model learns from emails labelled as spam or legitimate.
- Modern systems also use NLP and transformer models to detect subtle and evolving spam patterns.
- Important signals include message text, sender reputation, headers, links, attachments, and sending behavior.
- The business value is stronger protection against spam, phishing, and fraudulent communication.
- **In AT&T's context, this applies to workforce protection as well as network-scale filtering of spam, fraudulent messages, and robocalls**

Use Case 5: Topic Modelling of Movie Reviews

Domain: Entertainment

AI/ML Applied Technique: Unsupervised learning, Latent Dirichlet Allocation (LDA) topic modelling, Clustering, Deep Learning, Embedding-based topic models.

Data: Review-text corpus, star ratings, reviewer metadata, timestamps on timing of the reviews

Business Impact:

- Audience themes understood at scale
- Sentiment trends identified by topic
- Better content, marketing, and release decisions

Reasoning and Explanation:

- Topic modelling is an unsupervised-learning technique because there are no predefined labels.
- The model discovers recurring themes across large volumes of review text.
- Common themes may include plot, acting, pacing, visual effects, music, or overall sentiment.
- LDA and clustering can identify broad topic groups.
- Embedding-based models can capture deeper semantic meaning in the reviews.
- The business value is the ability to understand audience feedback at scale.
- **In AT&T's context, the same technique can be applied to customer reviews, app-store feedback, and care-call transcripts to detect emerging product and service issues early.**